

- 2 A person with diabetes has a blood glucose level that can be too high.

When the blood glucose level of a person without diabetes becomes too high, the liver stores glucose as a polysaccharide.

- (a) Which polysaccharide does the liver store?

(1)

- ☐ **A** amylopectin
- ☐ **B** cellulose
- ☐ **C** glycogen
- ☐ **D** starch

- (b) Blood glucose levels can become high following the digestion of carbohydrates.

Which of the following can be digested to release glucose?

(1)

- ☐ **A** both fructose and sucrose
- ☐ **B** both fructose and galactose
- ☐ **C** both galactose and lactose
- ☐ **D** both lactose and sucrose

- (c) Diabetes is a risk factor for cardiovascular disease.

- (i) One estimate is that there are 415 million people with diabetes in the world and that 46% of these people are undiagnosed.

Calculate the number of people who have undiagnosed diabetes.

(1)

Answer



- (ii) There are two types of diabetes, Type I and Type II.

The treatment for Type I diabetes is different from the treatment for Type II diabetes, so it is important for a correct diagnosis to be made.

Diagnosis can be difficult, particularly in people aged between 20 and 40 years old.

A genetic screening method is now available for the diagnosis of diabetes.

Explain why doctors are more likely to screen individuals once they develop diabetes than use methods such as prenatal testing.

(3)

(Total for Question 2 = 6 marks)

